MIL-P-51402(EA)
16 October 1975
(See section 6)

MILITARY SPECIFICATION

PLASTIC SHEET: POLYPROPYLENE-POLYETHYLENE LAMINATE FILM

This specification is approved for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 This specification covers one type of clear, transparent, flexible plastic sheet (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density.)
L-P-394 - Plastic Molding Material, Polypropylene, Injection and Extrusion.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipping (Civil Agencies).
FED-STD-191 - Textile Test Methods.
FED-STD-406 - Plastic, Methods of Testing.
FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.

FSC 9330

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D374-68 - Thickness of Solid Electrical Insulation, Tests for.
D2176-69 - Folding Endurance of Paper by the MIT Tester, Tests for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material. The plastic sheet shall be a clear, flexible, transparent, balanced, biaxially oriented polypropylene film entirely coated on one side with polyethylene. Polyethylene resin shall be in accordance with L-P-390, type 1, class 1, category 4 or 5, Natural Color. Polypropylene resin shall be in accordance with L-P-394, type IV, grade B, Natural Color.

3.1.1 Length. The sheet shall be furnished in continuous lengths of no less than 50 yards.

3.1.2 Width. Unless otherwise specified in the contract or purchase order the width of the sheet shall be 30.0 (+0.5 or -0.0) inches.

3.1.3 Thickness.

3.1.3.1 Laminate. The polypropylene-polyethylene laminate shall have an overall thickness of 0.00178 inch with a tolerance of plus or minus 10 percent when tested in accordance with 4.4.4.2.

3.1.3.2 Polypropylene. The polypropylene film thickness shall be 0.001 inch (+10 percent -0 percent) (see 4.1.3).

3.1.3.3 Polyethylene. The thickness of the polyethylene coating shall be 0.00078 inch with a tolerance of plus or minus 10 percent (see 4.1.3).

be cooled to room temperature

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3.2 Put-up.

3.2.1 Roll. The sheet shall be furnished in rolls containing no less than 50 yards.

3.2.2 Core. The sheet shall be furnished on cores with a length equal to or up to 1/8 inch greater than the width of the sheet. Unless otherwise specified (see 6.2) the inside diameter of the core shall be not less than 1-1/4 inches or more than 3-1/8 inches.

3.3 Physical characteristics. The sheet shall have the physical and agent resistant characteristics specified in table I.

Table I. Physical and agent resistant properties

Property	Requirement	Test
Tensile strength, psi minimum		
machine direction	7500	4.4.4.3
transverse direction	8000	4.4.4.3
Elongation, % minimum		
machine direction	40	4.4.4.3
transverse direction	40	4.4.4.3
Abrasion resistance, cycles minimum (Polypropylene side only)	90	4.4.4.3
Low temperature (20°F ± 2°F)	No visible cracking or damage	4.4.4.3
fold effect		
hydrostatic, PSI minimum	40	
Endurance of fold, cycles minimum	10,000	4.4.4.3
Slip (Kinetic coefficient of function)	0.5 maximum	4.4.4.5
Resistance to agent, minutes average		
H	75	4.4.4.3
GB	100	4.4.4.3

3.4 Bondability and strength of splice. The polyethylene side of the sheet shall be capable of being bonded to itself by commercially common, heat or electronic sealing devices and methods. The strength of such splices in either the machine or transverse direction of the sheet shall be no less than 12 pounds per inch width when tested in accordance with 4.4.4.4.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of sheet shall be produced in accordance with this specification for examination and tests (see 4.3).

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3.6 Workmanship. The sheet shall be clean, uniform in finish, free from dirt and other foreign matter, scratches, scuffs, cracks, greases, blisters, bubbles, pits, tears, cuts, holes, or other defects which may affect its appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government responsibility. The Government will be responsible for performing the H and GB agent resistance tests specified in 4.4.4.3.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of one roll of plastic sheet shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure. The preproduction sample of plastic sheet shall be examined and tested for conformance to all the requirements of this specification.

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4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the plastic sheet produced in no more than three consecutive work shifts in one week by one manufacturer, at one plant, using the same materials and processes, under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. The unit of product shall be one roll.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-1. Thirty square feet of plastic sheet shall be taken from the end of each sample roll.

4.4.3 Inspection provisions.

4.4.3.1 For examination and nondestructive tests. The sample plastic sheet shall be examined and tested in accordance with the classification of defects 4.4.3.3 and MIL-STD-105.

4.4.3.2 For destructive tests. The sample plastic sheet shall be tested in accordance with 4.4.4.3, 4.4.4.4 and 4.4.4.5. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Plastic sheet (polypropylene-polyethylene laminate).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Thickness	4.4.4.2
102	Width (3.2.1)	
103	Workmanship (3.6)	

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4.4.4 Tests.

4.4.4.1 Specimen preparation. No specimen shall be taken closer than 10 percent of width of film from either edge. Four test specimens for each test shall be tested unless otherwise specified in the method.

4.4.4.2 Thickness. The thickness of the sheet shall be determined in accordance with method C of ASTM D374.

4.4.4.3 Physical and agent resistance characteristic. The sheet shall be tested for the physical and agent resistance characteristics in accordance with the test methods specified in table II.

Table II. Physical and resistance to agents characteristics.

Characteristic	Test method
A. Physical test:	
Tensile strength	Method 1013, FED-STD-406
Percent elongations	Method 1013, FED-STD-406
Abrasion resistance (Polypropylene side only)	Method 5302, (Stoll), FED-STD-191 1/
Low temperature effect	Method 5874, FED-STD-191 2/
Endurance of fold	ASTM D2176
B. Resistance to toxic agents:	
H	Method 204, MIL-STD-282
GB	Method 206, MIL-STD-282

1/ The test shall be performed with an abradant of No. 1 sandpaper (Buehler 20), an air pressure of 3 psig, and an arm weight of 1 pound.

2/ Test method 5874 shall be modified to substitute method 10511 of FED-STD-601 in place of 3.2 of method 5874 of FED-STD-191.

4.4.4.4 Bondalibility and strength of splice. The strength of splice shall be determined in accordance with method 4311 of FED-STD-601.

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4.4.4.5 Slip.

4.4.4.5.1 Apparatus. The inclined plane apparatus shall consist of the following:

a. A smooth board approximately 26 by 26 by 3/4 inch to serve as the plane.

b. A steel block 2-1/2 by 3/4 inch to serve as a carrier and a loading medium for the sliding specimen.

c. A piece 12 by 2-3/4 by 1/8 inch thick cellular rubber sheeting of medium density as a cover for the block to assure intimate contact between the sliding and the stationary specimens. The rubber is brought up and around the ends of the block (sled) to fit snugly and is taped in place. The total weight of block and rubber shall be 1000 g.

d. An interval timer.

e. A vertical coefficient of friction indicator (tangent scale) graduated in 0.02 tangent units.

f. Four strips of cellophane tape 20 inches in length and 1 inch wide shall be taped approximately 4 inches apart to the top surface of the plane so that the long dimension is parallel to the sliding motion of the block. These strips are marked at intervals of 2.5 inches with a wax crayon to serve as the scale for the sliding block.

4.4.4.5.2 Specimens. Extreme care shall be taken in handling the material to keep the surface free of dust, lint, or fingerprints. Specimens for the plane shall be 22 inches in the machine direction and 17 inches in the transverse direction. The specimens for the sponge-rubber covered metal sled shall be 12 inches in the machine direction and 3 inches in the transverse direction. Both specimens shall be cut from adjacent areas of the sample piece. Sufficient block and plane specimens of each sample shall be cut so that tests of sides A to A, A to B, and B to B can be made.

4.4.4.5.3 Procedure. Level the apparatus. Tape the larger specimen to the clean, dry surface of the plane over the crayon-marked cellophane tape, smoothing the sheet sufficiently to eliminate wrinkles without stretching it. Tape the smaller specimen to the sponge-rubber covered metal block with the machine direction of the sheet parallel to the length of the block, pulling tightly to eliminate wrinkles without

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stretching the sheet. All combinations of the two sides, A and B of the sheet shall be tested. With the plane at an angle of zero degree, place the sheet covered block on the sheet specimen.

a. On the plane so that the long dimension is parallel to the inclination when the plane is tilted.

b. With the two polyethylene film surfaces in contact.

c. Near the end of the plane that will be elevated.

Slowly elevate the plane until that angle is reached which is just sufficient to allow the block to slide of its own accord. Stop elevating the plane at this point. If the sheet-covered block stops sliding before it is traveled 12.5 inches, change the sheet of the block and place it on a fresh portion of the sheet on the plane. Then elevate the plane in increments of 0.02 tangent units on the vertical coefficient of friction indicator, until the block travels, unaided, at least 12.5 inches. The tangent of the angle at which the block slides this distance is recorded as the static coefficient of friction of the specimen being tested. This tangent is read directly from the vertical coefficient of friction indicator, the top side of the plane being the reference point. Do not test a block specimen more than once for test purposes.

Raise the plane to a position 0.02 to 0.04 tangent units higher than the static coefficient of friction. Mount a new sheet specimen on the metal block and place the block on the plane for test as described above. Hold the block in place for at least ten seconds and then release. If the block does not begin to slide of its own accord, nudge it slightly to initiate sliding. When the block begins to slide, start the timer and determine the time it takes for the block to slide 2.5, 5.0, 7.5, 10.0, and 12.5 inches. The rate of sliding shall be approximately constant over the 12.5 inch length. Only the time for the last ten inches is used to calculate the coefficient of friction. If any marked accelerating or deceleration is noted, discard the determination and repeat the test. Calculate the reciprocal rate of sliding as follows:

$$\frac{\text{Time to slide 12.5 in. - time to slide 2.5 in.}}{10} = \text{rate in sec./in.}$$

This gives the rate at which the block slides down the plane 10 inches. Repeat this procedure at various angles using fresh specimens each time until at least two reciprocal rates above and two below 30 sec./in. are obtained. The kinetic coefficient of friction is calculated at a reciprocal rate of sliding of 30 sec./in. The rates of sliding above and

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below 30 sec./in. are plotted with rates of sliding versus the corresponding tangents. The tangent corresponding to a reciprocal rate of 30 sec./in. is taken from this plot. This tangent is numerically equal of the kinetic coefficient of friction of the sample. If a reciprocal rate of sliding slower than 30 sec./in. cannot be obtained, then the kinetic coefficient of friction is taken as the tangent of the lowest angle at which the block will slide at least 12.5 in.

5. PREPARATION FOR DELIVERY

5.1 Interplant shipment (see 6.3) The plastic sheet shall be packaged and packed to prevent damage in shipment from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123, and shall include the contract or purchase order number, and name and address of the prime contractor.

6. NOTES

6.1 Intended use. The plastic sheet covered by this specification is intended for use in the fabrication of overboots to be used in conjunction with various collective protection equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Width of the rolls.
(c) Inside diameter of core.
(d) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Interplant shipment. Packaging and packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a prime contractor, a subcontractor or a supplier to a military arsenal or plant.

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6.4 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-P-1081, dated 17 June 1971.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 9330-A637

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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